Radar range and velocity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate a radar's range measurement from its line-of-sight velocity measurement.

SUCCESS CRITERIA - I can...

1. Use round-trip time for range.
2. Explain the information in a Doppler shift.
3. State what radial velocity leaves unknown.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two EM waves in vacuum have frequencies f and 3 f. Compare speeds and wavelengths.

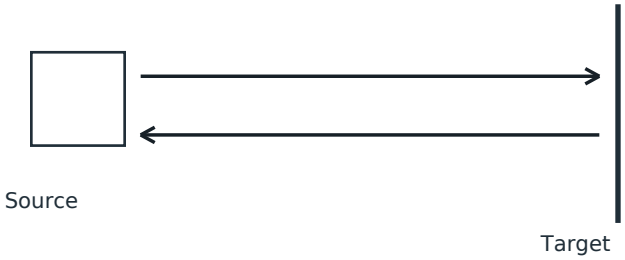
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: range = c Δt/2; c = 3.0×10^{8} m/s.
Radar time is a round trip. Doppler gives line-of-sight velocity information.

Worked reasoning: A radar echo returns after 2 microseconds. With $c=3\times10^{8}$ m/s, range=c Δt/2=300 m. The measured time includes the outward and return paths.

Your turn: Draw the round trip and label the conversion 2 microseconds=2×10^{-6} s.



Marzano check - circle: ① ② ③

03 Read and interpret the model

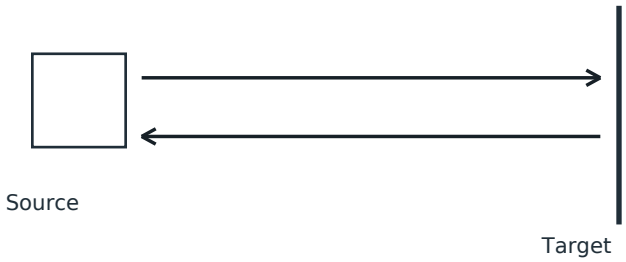
Draw outgoing and returning radar paths. Which measurement helps range: echo delay or frequency shift? Which helps radial velocity?



Marzano check - circle: ① ② ③

04 Apply the physics

A scaled radar animation uses signal speed 20 m/s and an echo delay of 3 s. Label the round trip. Find total travel distance and one-way range. This animation deliberately uses a slower speed than real radar.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

An asteroid has zero measured radial velocity at one instant. Does that prove it is stationary? Sketch another possible motion.

Add a velocity perpendicular to the line of sight.

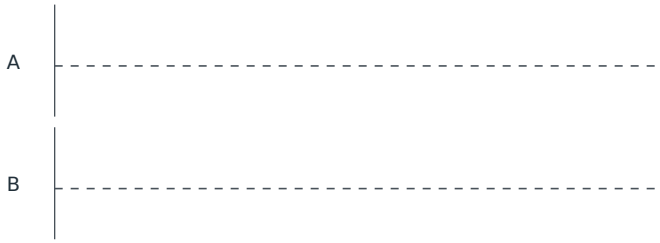


Observer Target

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A wave has frequency 5 Hz and wavelength 7 m. Find its speed.
Label a wave sketch to support the calculation or comparison.

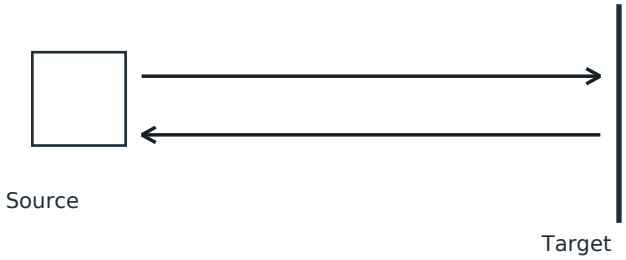


Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

An echo’s out-and-back distance represents 1000 m. How far is the target?



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Echo times change from 4 to 3 microseconds. Calculate both ranges and state whether the target is approaching or receding.



Marzano check - circle: 1 2 3